Name:_____ Period:_____ **Assignment: Introductory ESS Language Decoder**

**Activity 1: Terms Relating to Dimensions and Scale**

Scientists often need to describe where something happens, when it happens, how things compare, estimate values, and determine the order of events. In this activity, you will use the story and map of the Ever Laurel accident to explore five important scientific terms:

**SPATIAL • RELATIVE • APPROXIMATE • TEMPORAL• SEQUENCE**

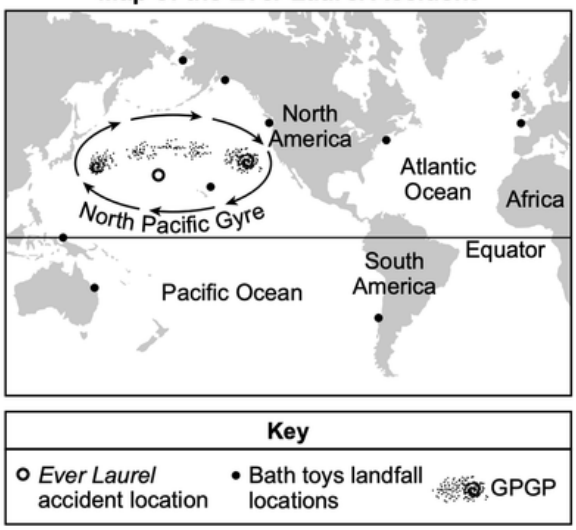
**Rubber Ducks and the Pacific Ocean Trash**

On January 10, 1992, the container ship Ever Laurel lost twelve 40-foot containers during a storm at sea. One container broke open and released over 28,000 plastic bath toys into the ocean. These bath toys floated on the ocean surface for years and made landfall on beaches worldwide.

The Pacific Ocean is home to the Great Pacific Garbage Patch (GPGP). This island of plastic is 1.6 million square km in size, approximately 11 times the size of New York State.

The map below contains information about the Ever Laurel accident, locations where bath toys were found, and the Great Pacific Garbage Patch.

**Map of the *Ever Laurel* Accident**




➤**SPATIAL: Used to describe changes, differences, or patterns in physical space or geographic location.**

Scientists use the word spatial when they are interested in the location of something or how locations are arranged across an area.

🤔 *Think: Where?*

Use the Map of the Ever Laurel Accident to answer the questions below. The map shows the accident location, bath-toy landfall locations, and the Great Pacific Garbage Patch.

1. In which ocean did the Ever Laurel accident occur? _____

2. Look at the locations where the bath toys eventually made landfall.
   What do you notice about WHERE these locations are found? _____

3. The locations where the bath toys made landfall provide spatial information because they tell us:
   □ when the toys were found. □ how long the toys traveled
   □ what order events occurred. □ where the toys were found

➤**RELATIVE: Used to compare one value, amount, size, or position to another.**

Scientists use relative when they describe something by comparing it to something else. A relative description does not necessarily give an exact measurement.

🤔 *Think: Compared to what?*

Look again at the Ever Laurel map.

1. Describe the location of the Great Pacific Garbage Patch relative to the Ever Laurel accident location.

   The Great Pacific Garbage Patch is located _____

2. Why is the word relative appropriate when describing these locations?
   □ We are determining the exact location of each place. □ We are determining when each event occurred.
   □ We are comparing the position of one location to another. □ We are estimating the distance between the locations.

➤**APPROXIMATE / APPROXIMATELY: Indicates an estimated value, amount, distance, or time that is close to, but not exactly, the actual value.**

🤔*Think: About how much?*

1. According to the passage, approximately how many times larger than New York State is the Great Pacific Garbage Patch? Approximately _____ times larger

2. Which statement best explains what approximately 11 times means?
   □ It is exactly 11 times larger with no difference at all. □ It is estimated to be about 11 times larger.
   □ Scientists do not know anything about its size. □ It must be more than 11 times larger.

3. A scientist tells you that an object traveled approximately 500 kilometers. Which would be a reasonable actual distance?
   □ 2 km □ 498 km
   □ 5,000 km □ exactly 500 km only

4. Why might scientists sometimes use an approximate value rather than an exact value?_____

   _____

➤***TEMPORAL: Refers to time, changes over time, or the timing of specific events.***

While spatial information tells scientists where something happens, temporal information tells scientists when something happens or how something changes with time.

🤔*Think: When? / How long? / Over time?*

1. What piece of temporal information tells us when the accident occurred? _____

2. What piece of temporal information tells us about how long the toys remained in the ocean?_____

3. Which question is asking for TEMPORAL information?
   □ Where did the bath toys make landfall? □ Which ocean contains the Great Pacific Garbage Patch?
   □ When did the Ever Laurel accident occur? □ Where is the garbage patch located?

4.Spatial or Temporal? Write S for spatial or T for temporal.
   ____ The accident occurred on January 10, 1992.
   ____ The accident occurred in the Pacific Ocean.
   ____ The toys floated on the ocean surface for years.
   ____ Bath toys made landfall on beaches worldwide.

➤**SEQUENCE: The particular order in which a series of events or processes occurs.**

Scientists often need to determine the sequence of events to understand how something happened.

🤔*Think: What order?*

1. The events below are out of order. Number them 1–5 to show the correct sequence.
   ____ The plastic bath toys floated on the ocean surface.
   ____ One of the containers broke open.
   ____ Bath toys made landfall on beaches around the world.
   ____ The Ever Laurel lost containers during a storm.
   ____ More than 28,000 plastic bath toys were released into the ocean.

2. Why is knowing the sequence different from simply knowing when an event happened?_____

   _____

**Activity 2: Terms Relating to Scientific Models & Data Analysis**

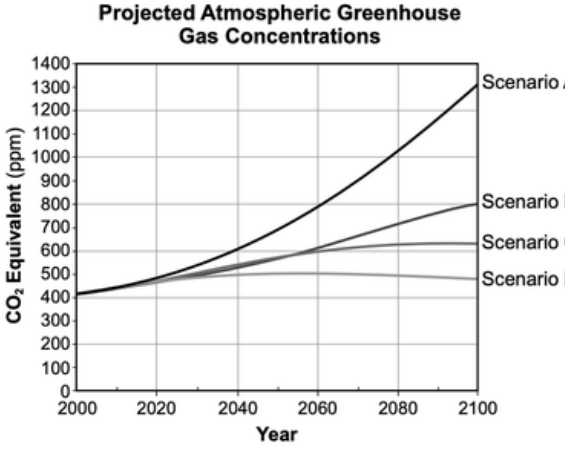
Scientists collect information, organize data, and use it to understand how the world works. To do this successfully, you need to understand the language scientists use when they represent information, analyze data, notice trends, and support ideas. In this activity, you will investigate graphs, maps, diagrams, and data to practice eight important terms:

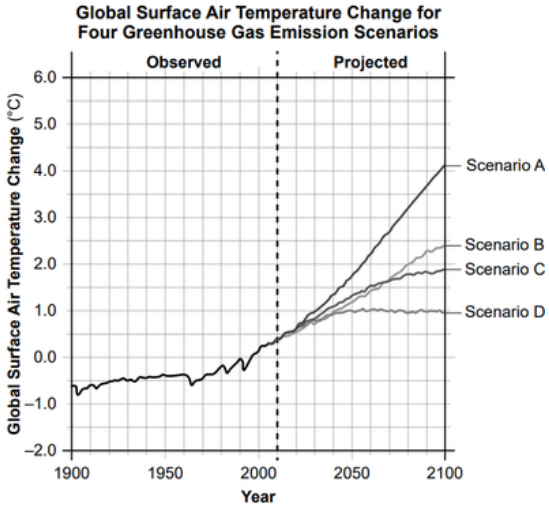
**MODEL • PATTERN • RELATIONSHIP • PROJECTED • DISTRIBUTION • ANOMALY • VARIABLE • FACTOR • CLAIM • EVIDENCE**

**Part 1: Looking into the Future**

Computer-based global climate models are helpful tools for collecting data on projected future climate conditions. These models use various scenarios, or possibilities, that assume different human-based decisions on how we address greenhouse gas emissions.

The graph below shows future greenhouse gas concentrations for four different greenhouse gas emission scenarios measured in parts per million (ppm)




➤ **MODEL: A representation of an object, process, or system that scientists use to understand, explain, or predict how something works.**

🤔*Think: How is it being represented?*

1. In this investigation, what are scientists using models to study?_____

2. Why might scientists use a model when studying what Earth's climate could be like in the future?_____

_____

➤ **PATTERN: A recognizable trend, repetition, or regular way that something happens or changes.**

🤔*Think: What trend do I notice?*

Examine the graph showing projected atmospheric greenhouse gas concentrations.

1. Describe the overall pattern in greenhouse gas concentration for Scenario A from 2000 to 2100._____

_____

2. Does every scenario show the same pattern? Explain one difference you notice._____

_____

➤ **RELATIONSHIP: The way two or more variables are connected or how they change in relation to one another.**

🤔*Think: How are they connected?*

Now compare the greenhouse gas concentration graph with the global surface temperature graph. The Regents provides both projected greenhouse-gas concentrations and projected global-temperature changes for the same four scenarios.

1. What happens to projected global surface temperature as projected greenhouse gas concentration increases?

As greenhouse gas concentration _____, global surface temperature generally _____.

➤**PROJECTED: Predicted or estimated to happen in the future based on available information or data.**
🤔*Think: What is expected to happen in the future?*

The global surface temperature graph contains both observed and projected information.

1. Which word describes data that have already been measured? □ Observed □ Projected

2. Which word describes data that scientists predict may occur in the future? □ Observed □ Projected

3. Are projected values guaranteed to happen exactly as shown? □ Yes □ No

4. What information are the scientists using to make projections about future climate conditions?_____

_____

➤**DISTRIBUTION: The way values or information are spread out across an area, range, or set of categories.**
🤔*Think: How are the values spread out?*

Look at the projected greenhouse gas concentrations for Scenarios A, B, C, and D in the year 2100.

1. Which scenario has the highest projected value?_____

2. Which scenario has the lowest projected value?_____

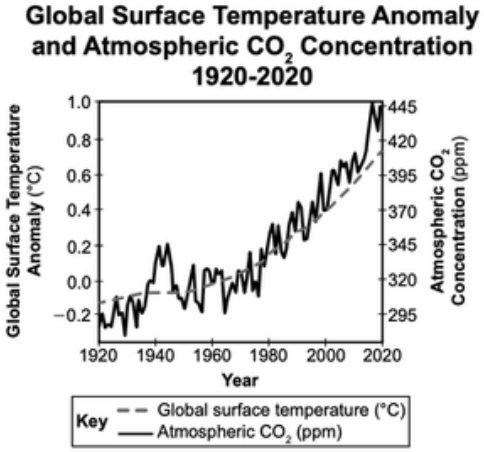
3. Are the projected values in 2100 clustered closely together or spread across a wide range?_____

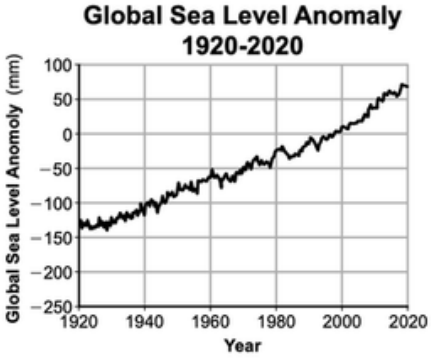
4. Based on your answer above, describe the distribution of the projected values._____

_____

**Part 2: Looking at the Past**

The graphs below show some information about global temperature, CO2, and sea level. An anomaly is a change from an expected value. An anomaly value of 0 represents no change from the historical average.




➤**ANOMALY: A difference or change from an expected or historical average value.**
🤔*Think: How different is it from the expected or average value?*

1. An anomaly value of 0°C means:
   □ The temperature was 0°C.
   □ There was no change from the historical average.
   □ The temperature decreased.
   □ No temperature data were collected.

2. What would a positive temperature anomaly tell you?_____

3. What would a negative temperature anomaly tell you?_____

4. Examine the graph. How did the global surface temperature anomaly change from 1920 to 2020?_____

_____

➤**VARIABLE: A factor, condition, or quantity that can change or have different values in an investigation, model, or data set.**

Scientists study variables to determine how changes in one may be related to changes in another.

🤔*Think: What can change?*

Look at the two climate graphs. They show several variables that change over time, including atmospheric greenhouse gas concentration and global surface air temperature.

1. Identify two variables shown in the climate graphs. Variable 1: _____ Variable 2: _____

2. How do you know these are variables?
   □ They stay exactly the same. □ They can change or have different values.
   □ They are predictions only. □ They are units of measurement.

3. Complete the statement: A variable is something that can _____.

➤**FACTOR: Something that contributes to, influences, or affects a result or outcome.**

Scientists often investigate different factors that may cause or influence changes in a system.

🤔*Think: What could affect it?*

The climate models use different scenarios based on human decisions about greenhouse gas emissions.

1. According to the climate-model information, identify one factor that could affect future climate conditions._____

_____

2. Why would scientists consider multiple factors when creating a model of future conditions?_____

_____

3. Which statement is describing a FACTOR?
   □ Global surface temperature was measured in degrees Celsius. □ The graph contains data from 2000 to 2100.
   □ Greenhouse gas emissions can influence future climate conditions.□ Scenario A has the highest value in 2100.

➤**CLAIM + EVIDENCE** (These two words work together.)

**CLAIM: A statement or conclusion that something is true and can be supported with evidence.**

🤔*Think: What am I saying is true?*

**EVIDENCE: Information, observations, or data used to support a claim or explanation.**

🤔*Think: How do I know?*

A student makes the following claim: Earth's global surface temperature increased from 1920 to 2020.

1. Does the graph support this claim? □ Yes. □ No

2. Identify one piece of EVIDENCE from the graph that supports your answer. Be specific!_____

_____

3. Which of the following is the CLAIM and which is the EVIDENCE? *Write C for Claim or E for Evidence.*
   ____ A. Global surface temperature has increased over time.
   ____ B. The temperature anomaly increased from approximately _____°C in 1920 to approximately _____°C in 2020.

**Activity 3: Terms Relating to Features, Systems, and Engineering**

Scientists and engineers don't just study how things work. They also identify problems, examine how different parts of a system interact, and design solutions. A solution may work well in one way but create a problem in another, so scientists and engineers must consider benefits, limitations, and possible effects. In this activity, you will investigate a real engineering problem while learning important scientific terms:

**FEATURE • SYSTEM • COMPONENT • CRITERIA • CONSTRAINT • TRADE-OFF**
**MITIGATION • FEEDBACK • INPUT • OUTPUT • SIMULATION • SOLUTION**

Managing and mitigating lava flows from volcanic eruptions is a challenge due to the intense heat and fluid dynamics of lava. In Iceland, several solutions have been applied or proposed to address the impacts of lava flows.

The table below lists different methods that all require equipment to address the impact of lava fl ows on communities.

**Possible Lava Flow Solutions**

| | Solution | Description | Constraint |
|---|---|---|---|
| A | Lava diversion | Construct channels to redirect lava | – Costly<br>– Time-consuming<br>– Lava may breach<br>– Redirects flows instead of stopping them |
| B | Seawater cooling | Use pumps to draw seawater to spray lava to cool and harden | – Need constant water supply<br>– Demand for energy to run multiple costly pumps |
| C | Lava flow barrier | Build earthen barriers to dam the lava and stop its flow | – Need machinery/equipment to build barrier<br>– Lava could breach the barrier |
| D | Explosions | Detonate explosives to disrupt the lava flow and redirect its path | – Could cause further damage and loss of life<br>– Not precise and lava returns to its original path |

**Part 1: Understanding the Problem**

➤**FEATURE: A distinctive or noticeable characteristic, part, or aspect of something.**

🤔 *Think: What characteristic or part do I notice?*

Look at the four proposed lava-flow solutions.

1. Identify one feature described in Solution C *(there can be multiple answers).*_____

2. Identify one feature that makes Solution B different from Solution C._____

➤**SYSTEM: A group of interacting or connected parts that work together as a whole.**

🤔 *Think: What parts are working together?*

A community affected by a volcanic eruption can be viewed as part of a larger system. Lava, land, people, buildings, equipment, and the proposed solutions can affect one another.

1. Why can this situation be considered a system rather than a collection of completely unrelated parts?_____

_____

➤**COMPONENT: One individual part of a larger system.**

🤔 *Think: What is one part of the whole?*

1. Check THREE components that could be part of the lava-flow management system.
   □ Lava flow □ Pumps □ Community □ Moon □ Barriers □ Jupiter

2. Choose one component you checked. Explain how it interacts with another component of the system._____

_____

**Part 2: Designing a Solution**

**➤SOLUTION: A method or action designed to solve a problem or address a challenge.**

🤔*Think: How could the problem be addressed?*

Four possible solutions are listed in the data table.

1. What problem are all four solutions attempting to address?_____

2. Which solution do you initially think would work best? Solution _____ Why? Justify using information from the table.

_____

**➤CRITERIA: The requirements or desired characteristics that a successful solution should meet.**

🤔*Think: What does the solution need to do?*

Engineers establish the following criteria for a successful solution:

- Protect homes and people
- Reduce lava damage
- Be reliable during an eruption

1. Which proposed solution BEST meets all three criteria? Solution _____

2. Use information from the table to explain why your chosen solution meets the criteria._____

_____

**➤CONSTRAINT: A limitation or restriction that must be considered when designing or choosing a solution.**

🤔*Think: What limits my options?*

The data table identifies several constraints, including cost, time, equipment, water supply, energy use, and safety.

1. Identify one constraint of using seawater cooling._____

2. Identify one constraint of using explosions._____

3. Why must engineers consider constraints before selecting a solution?_____

_____

**➤TRADE-OFF: A compromise in which gaining a benefit in one area may result in a disadvantage or cost in another.**

🤔*Think: What do I gain, and what do I give up?*

Most engineering solutions are not perfect. A solution may solve one problem while creating another. For example, seawater cooling could help cool and harden lava, BUT it requires a constant water supply and energy to operate costly pumps.

1. Choose ONE proposed solution and identify the benefit and trade-off. Solution: _____

   Benefit: _____

   Trade-off: _____

2. Why might engineers choose a solution even if it has a negative trade-off?_____

_____

**➤MITIGATION: An action taken to reduce the severity, impact, or harmful effects of a problem.**

🤔*Think: How can the impact be reduced?*

Notice something important: engineers may not be able to stop the volcanic eruption itself. Instead, they can try to reduce the damage it causes.

1. Explain why redirecting a lava flow is an example of mitigation rather than preventing the volcanic eruption._____

_____

**Part 3: Testing a Solution**

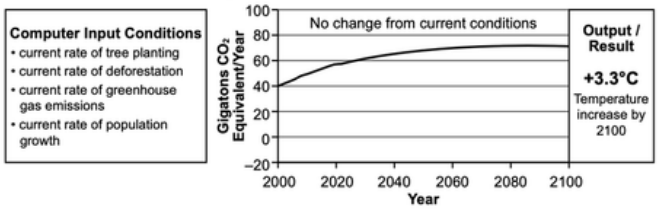
**Human Impact on Earth**

Global human population has increased dramatically in the last 50 years, which has increasingly stressed Earth's natural resources. This has created challenges in the management of those natural resources while also creating opportunities to find solutions to those challenges.

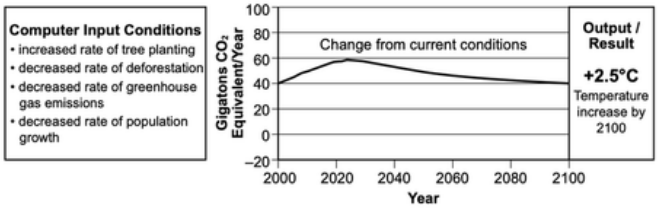
The EN-ROADS simulator is an online tool that allows users to manipulate variables in order to see their effects on climate change. A student used this simulator to estimate the global temperature change by the year 2100 if no changes to current conditions were made and if changes from current conditions were made.

The graphs show the results of the simulations. Simulation 1 shows the possible outcome of greenhouse gas net emissions and the associated atmospheric global temperature change under current conditions. Simulation 2 shows the possible outcome of greenhouse gas net emissions and the associated atmospheric global temperature change due to a different set of computer input conditions.

**Simulation 1: Greenhouse Gas Net Emissions– No Change from Current Human Practices**




**Simulation 2: Greenhouse Gas Net Emissions– Changed Human Practices**




➤**SIMULATION: An imitation or model of a real-world situation or process used to investigate what may happen under different conditions.**

*Think: What happens if..?*

The EN-ROADS simulator allows users to change variables and observe possible effects on climate conditions.

1. Why might scientists use a simulation instead of testing these changes on the actual Earth?_____

_____

2. What can scientists learn by changing conditions in a simulation?_____

➤**INPUT & OUTPUT** (These two words work together.)

**INPUT: Information, conditions, matter, or energy that enters a system or is put into a model.**

*Think: What goes in?*

**OUTPUT: The information, result, matter, or energy that comes out of a system or model.**

*Think: What comes out?*

The simulation actually labels these Computer Input Conditions and Output/Result. Investigate Simulation 2.

1. Circle ALL of the inputs listed in Simulation 2. Check this box when you have done it. □

2. Why are these called inputs?_____

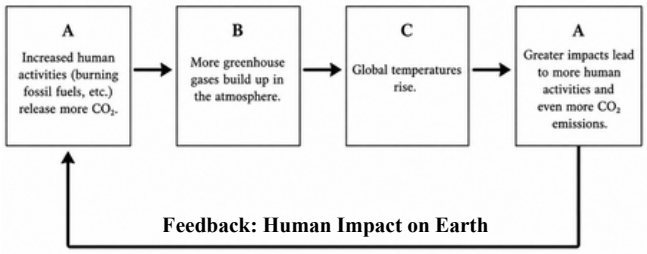
3. Complete the statement using input and output: Scientists changed the _____ and observed the resulting _____.

➤**FEEDBACK: A process in which a change in one part of a system causes effects that can influence other parts of the system, including the original change.**

*Think: How can one change lead to more change?*

1. What makes feedback different from a simple cause-and-effect relationship? Use the example to the right._____

_____




2. In a feedback, the result of a change can:

□ have no effect on the system. □ influence further changes within the system

□ only occur in computer models □ always stop the original change